

Introduction to Optimization
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University of Science and Technology of China

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Homework 3
Due: Apr. 13th, 2025
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Exercise 1:

Derive the dual of the following LPs. Also compute the dual of the dual for each case and interpret your findings.

(a)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax \succeq b\end{array}$$

(b)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax \succeq b, x \succeq 0\end{array}$$

Solution 1:

(a)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \succeq b \end{array} \iff \begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & b - Ax \preceq 0 \end{array}$$

Introduce the multiplier $\lambda \succeq 0$. The Lagrangian is

$$L(x, \lambda) = c^T x + \lambda^T (b - Ax).$$

Expanding and rearranging,

$$L(x, \lambda) = c^T x + b^T \lambda - \lambda^T Ax = b^T \lambda + (c - A^T \lambda)^T x.$$

Therefore the dual function is

$$g(\lambda) = \inf_x L(x, \lambda) = \inf_x \left[b^T \lambda + (c - A^T \lambda)^T x \right].$$

If $c - A^T \lambda = 0$, then the term depending on x vanishes, so

$$g(\lambda) = b^T \lambda.$$

If $c - A^T \lambda \neq 0$, then

$$\inf_x (c - A^T \lambda)^T x = -\infty,$$

hence

$$g(\lambda) = -\infty.$$

Thus

$$g(\lambda) = \begin{cases} b^T \lambda, & A^T \lambda = c, \\ -\infty, & \text{otherwise.} \end{cases}$$

Hence the dual problem is

$$\begin{array}{ll} \text{maximize} & b^T \lambda \\ \text{subject to} & A^T \lambda = c, \\ & \lambda \succeq 0. \end{array}$$

Next we compute the dual of this dual. Rewrite the dual as the equivalent minimization problem

$$\begin{array}{ll} \text{minimize} & -b^T \lambda \\ \text{subject to} & A^T \lambda - c = 0, \\ & -\lambda \preceq 0. \end{array}$$

Let z be the multiplier for the equality constraint $A^T \lambda - c = 0$, and let $\mu \succeq 0$ be the multiplier for $-\lambda \preceq 0$. Its Lagrangian is

$$L(\lambda, z, \mu) = -b^T \lambda + z^T (A^T \lambda - c) + \mu^T (-\lambda).$$

Rearranging,

$$L(\lambda, z, \mu) = -b^T \lambda + \lambda^T A z - c^T z - \mu^T \lambda = \lambda^T (A z - b - \mu) - c^T z.$$

Hence the dual function is

$$g(z, \mu) = \inf_{\lambda} L(\lambda, z, \mu).$$

If $Az - b - \mu = 0$, then

$$g(z, \mu) = -c^T z;$$

otherwise,

$$g(z, \mu) = -\infty.$$

Therefore

$$g(z, \mu) = \begin{cases} -c^T z, & Az - b - \mu = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

So the dual of the dual is

$$\begin{aligned} & \text{maximize} && -c^T z \\ & \text{subject to} && Az - b - \mu = 0, \\ & && \mu \succeq 0. \end{aligned}$$

Eliminating μ , we obtain

$$Az - b \succeq 0 \iff Az \succeq b.$$

Hence the dual of the dual is equivalently

$$\begin{aligned} & \text{maximize} && -c^T z \\ & \text{subject to} && Az \succeq b, \end{aligned}$$

or, equivalently,

$$\begin{aligned} & \text{minimize} && c^T z \\ & \text{subject to} && Az \succeq b. \end{aligned}$$

This is exactly the original primal problem, up to renaming the variable.

Interpretation: in part (a), the primal variable x is free in sign, and this corresponds to the equality constraint $A^T \lambda = c$ in the dual.

(b) Now consider

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax \succeq b, \quad x \succeq 0. \end{aligned}$$

Rewrite the constraints as

$$b - Ax \preceq 0, \quad -x \preceq 0.$$

Introduce multipliers $\lambda \succeq 0$ for $b - Ax \preceq 0$ and $\mu \succeq 0$ for $-x \preceq 0$. The Lagrangian is

$$L(x, \lambda, \mu) = c^T x + \lambda^T (b - Ax) + \mu^T (-x).$$

Rearranging gives

$$L(x, \lambda, \mu) = b^T \lambda + (c - A^T \lambda - \mu)^T x.$$

Therefore

$$g(\lambda, \mu) = \inf_x L(x, \lambda, \mu) = \inf_x \left[b^T \lambda + (c - A^T \lambda - \mu)^T x \right].$$

If

$$c - A^T \lambda - \mu = 0,$$

then

$$g(\lambda, \mu) = b^T \lambda.$$

Otherwise,

$$g(\lambda, \mu) = -\infty.$$

Hence

$$g(\lambda, \mu) = \begin{cases} b^T \lambda, & c - A^T \lambda - \mu = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

So the dual problem is

$$\begin{aligned} & \text{maximize} && b^T \lambda \\ & \text{subject to} && c - A^T \lambda - \mu = 0, \\ & && \lambda \succeq 0, \mu \succeq 0. \end{aligned}$$

Eliminating μ , we get

$$\mu = c - A^T \lambda \succeq 0 \iff A^T \lambda \preceq c.$$

Therefore the dual is

$$\begin{aligned} & \text{maximize} && b^T \lambda \\ & \text{subject to} && A^T \lambda \preceq c, \\ & && \lambda \succeq 0. \end{aligned}$$

Next compute the dual of this dual. Rewrite it as

$$\begin{aligned} & \text{minimize} && -b^T \lambda \\ & \text{subject to} && A^T \lambda - c \preceq 0, \\ & && -\lambda \preceq 0. \end{aligned}$$

Let $z \succeq 0$ be the multiplier for $A^T \lambda - c \preceq 0$, and let $\mu \succeq 0$ be the multiplier for $-\lambda \preceq 0$. The Lagrangian is

$$L(\lambda, z, \mu) = -b^T \lambda + z^T (A^T \lambda - c) + \mu^T (-\lambda).$$

Rearranging,

$$L(\lambda, z, \mu) = -b^T \lambda + \lambda^T A z - c^T z - \mu^T \lambda = \lambda^T (A z - b - \mu) - c^T z.$$

Hence the dual function is

$$g(z, \mu) = \inf_{\lambda} L(\lambda, z, \mu).$$

If $Az - b - \mu = 0$, then

$$g(z, \mu) = -c^T z;$$

otherwise,

$$g(z, \mu) = -\infty.$$

Thus

$$g(z, \mu) = \begin{cases} -c^T z, & Az - b - \mu = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

Therefore the dual of the dual is

$$\begin{aligned} & \text{maximize} && -c^T z \\ & \text{subject to} && Az - b - \mu = 0, \\ & && z \succeq 0, \mu \succeq 0. \end{aligned}$$

Eliminating μ gives

$$Az - b \succeq 0 \iff Az \succeq b.$$

Hence the dual of the dual is equivalently

$$\begin{aligned} & \text{maximize} && -c^T z \\ & \text{subject to} && Az \succeq b, \\ & && z \succeq 0, \end{aligned}$$

or

$$\begin{aligned} & \text{minimize} && c^T z \\ & \text{subject to} && Az \succeq b, \quad z \succeq 0. \end{aligned}$$

This is again exactly the original primal problem, up to renaming the variable.

Interpretation: compared with part (a), the additional primal constraint $x \succeq 0$ changes the dual condition from

$$A^T \lambda = c$$

to

$$A^T \lambda \preceq c.$$

In other words, a free primal variable corresponds to an equality constraint in the dual, while a nonnegative primal variable corresponds to an inequality constraint in the dual.

In both cases, the dual of the dual recovers the original primal LP. ■

Exercise 2:

Lagrangian relaxation of Boolean LP. A Boolean linear program is an optimization problem of the form

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax \preceq b \\ & && x_i \in \{0, 1\}, \quad i = 1, \dots, n, \end{aligned}$$

and is, in general, very difficult to solve. In exercise 4.15 we studied the LP relaxation of this problem,

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax \preceq b \\ & && 0 \leq x_i \leq 1, \quad i = 1, \dots, n, \end{aligned} \tag{5.107}$$

which is far easier to solve, and gives a lower bound on the optimal value of the Boolean LP. In this problem we derive another lower bound for the Boolean LP, and work out the relation between the two lower bounds.

- (a) *Lagrangian relaxation.* The Boolean LP can be reformulated as the problem

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax \preceq b \\ & && x_i(1 - x_i) = 0, \quad i = 1, \dots, n, \end{aligned}$$

which has quadratic equality constraints. Find the Lagrange dual of this problem. The optimal value of the dual problem (which is convex) gives a lower bound on the optimal value of the Boolean LP. This method of finding a lower bound on the optimal value is called *Lagrangian relaxation*.

- (b) Show that the lower bound obtained via Lagrangian relaxation, and via the LP relaxation (5.107), are the same.

Hint: Derive the dual of the LP relaxation (5.107).

Solution 2:

(a) The Boolean constraints $x_i \in \{0, 1\}$ can be rewritten as

$$x_i(1 - x_i) = 0, \quad i = 1, \dots, n.$$

Hence the Boolean LP is equivalent to

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax \preceq b, \\ & && x_i(1 - x_i) = 0, \quad i = 1, \dots, n. \end{aligned}$$

Write the inequality constraint as

$$Ax - b \preceq 0.$$

Let $\lambda \succeq 0$ be the multiplier for $Ax - b \preceq 0$, and let $\nu \in \mathbb{R}^n$ be the multiplier for the equality constraints $x_i(1 - x_i) = 0$. The Lagrangian is

$$L(x, \lambda, \nu) = c^T x + \lambda^T (Ax - b) + \sum_{i=1}^n \nu_i (x_i - x_i^2).$$

Expanding and collecting terms gives

$$L(x, \lambda, \nu) = -b^T \lambda + (c + A^T \lambda)^T x + \sum_{i=1}^n \nu_i x_i - \sum_{i=1}^n \nu_i x_i^2.$$

Define

$$s_i := c_i + (A^T \lambda)_i.$$

Then

$$L(x, \lambda, \nu) = -b^T \lambda + \sum_{i=1}^n \left[-\nu_i x_i^2 + (s_i + \nu_i) x_i \right].$$

Therefore the dual function is

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = -b^T \lambda + \sum_{i=1}^n \inf_{x_i \in \mathbb{R}} \left[-\nu_i x_i^2 + (s_i + \nu_i) x_i \right].$$

For a single coordinate, define

$$\phi(s, \nu) := \inf_{x \in \mathbb{R}} (-\nu x^2 + (s + \nu)x).$$

We now compute $\phi(s, \nu)$ by cases.

If $\nu > 0$, then $-\nu x^2 + (s + \nu)x$ is concave in x , so

$$\phi(s, \nu) = -\infty.$$

If $\nu = 0$, then

$$-\nu x^2 + (s + \nu)x = sx.$$

Hence

$$\phi(s, 0) = \begin{cases} 0, & s = 0, \\ -\infty, & s \neq 0. \end{cases}$$

If $\nu < 0$, then the function is strictly convex in x , and

$$\phi(s, \nu) = -\frac{(s + \nu)^2}{4(-\nu)} = \frac{(s + \nu)^2}{4\nu}.$$

Thus

$$\phi(s, \nu) = \begin{cases} \frac{(s + \nu)^2}{4\nu}, & \nu < 0, \\ 0, & \nu = 0, s = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

Hence the dual function is

$$g(\lambda, \nu) = -b^T \lambda + \sum_{i=1}^n \phi(s_i, \nu_i), \quad s_i = c_i + (A^T \lambda)_i.$$

Therefore the Lagrange dual problem is

$$\begin{aligned} & \text{maximize} && -b^T \lambda + \sum_{i=1}^n \phi(c_i + (A^T \lambda)_i, \nu_i) \\ & \text{subject to} && \lambda \succeq 0, \end{aligned}$$

where ϕ is given above.

We can simplify this dual by maximizing over each ν_i . For fixed s , and $\nu < 0$,

$$\phi(s, \nu) = \frac{(s + \nu)^2}{4\nu} = \frac{s^2}{4\nu} + \frac{s}{2} + \frac{\nu}{4}.$$

Differentiating with respect to ν gives

$$\frac{\partial \phi}{\partial \nu} = -\frac{s^2}{4\nu^2} + \frac{1}{4}.$$

Setting this equal to zero, we obtain

$$-\frac{s^2}{4\nu^2} + \frac{1}{4} = 0 \iff \nu^2 = s^2.$$

Since we must have $\nu < 0$, the maximizer is

$$\nu^* = -|s|.$$

Substituting back,

$$\phi(s, \nu^*) = -\frac{|s|}{4} + \frac{s}{2} - \frac{|s|}{4} = \frac{s - |s|}{2} = \min\{0, s\}.$$

Therefore

$$\sup_{\nu \in \mathbb{R}} \phi(s, \nu) = \min\{0, s\}.$$

It follows that the dual can be written in the reduced form

$$\begin{array}{ll} \text{maximize} & -b^T \lambda + \sum_{i=1}^n \min\{0, c_i + (A^T \lambda)_i\} \\ \text{subject to} & \lambda \succeq 0. \end{array}$$

By weak duality, this optimal value is a lower bound on the optimal value of the Boolean LP.

(b) The LP relaxation is

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \preceq b, \\ & 0 \leq x_i \leq 1, \quad i = 1, \dots, n. \end{array} \quad (5.107)$$

Rewrite the constraints as

$$Ax - b \preceq 0, \quad -x \preceq 0, \quad x - \mathbf{1} \preceq 0.$$

Let the corresponding multipliers be

$$\lambda \succeq 0, \quad \mu \succeq 0, \quad \eta \succeq 0.$$

The Lagrangian is

$$L(x, \lambda, \mu, \eta) = c^T x + \lambda^T (Ax - b) + \mu^T (-x) + \eta^T (x - \mathbf{1}).$$

Expanding gives

$$L(x, \lambda, \mu, \eta) = -b^T \lambda - \mathbf{1}^T \eta + (c + A^T \lambda - \mu + \eta)^T x.$$

Therefore the dual function is finite if and only if

$$c + A^T \lambda - \mu + \eta = 0.$$

In that case,

$$g(\lambda, \mu, \eta) = -b^T \lambda - \mathbf{1}^T \eta.$$

So the dual of the LP relaxation is

$$\begin{aligned} & \text{maximize} && -b^T \lambda - \mathbf{1}^T \eta \\ & \text{subject to} && c + A^T \lambda - \mu + \eta = 0, \\ & && \lambda, \mu, \eta \succeq 0. \end{aligned}$$

Eliminate μ using

$$\mu = c + A^T \lambda + \eta.$$

Since $\mu \succeq 0$, this is equivalent to

$$c + A^T \lambda + \eta \succeq 0.$$

Hence the dual is equivalently

$$\begin{aligned} & \text{maximize} && -b^T \lambda - \mathbf{1}^T \eta \\ & \text{subject to} && c + A^T \lambda + \eta \succeq 0, \\ & && \lambda, \eta \succeq 0. \end{aligned}$$

For fixed λ , the objective is maximized by choosing the smallest feasible η , namely

$$\eta_i^* = \max\{0, -c_i - (A^T \lambda)_i\}.$$

Substituting this into the objective gives

$$-b^T \lambda - \sum_{i=1}^n \max\{0, -c_i - (A^T \lambda)_i\}.$$

Using

$$-\max\{0, -s\} = \min\{0, s\},$$

we obtain the reduced dual

$$\begin{aligned} & \text{maximize} && -b^T \lambda + \sum_{i=1}^n \min\{0, c_i + (A^T \lambda)_i\} \\ & \text{subject to} && \lambda \succeq 0. \end{aligned}$$

This is exactly the same problem as the reduced Lagrange dual obtained in part (a). Therefore the two lower bounds are the same.

Since the LP relaxation is a linear program, its optimal value equals the optimal value of its dual. Hence the lower bound obtained from the LP relaxation coincides with the lower bound obtained from the Lagrangian relaxation.

■

Exercise 3:

Consider the problem

$$\begin{array}{ll}\text{minimize} & x_1 \\ \text{subject to} & \sqrt{x_1^2 + x_2^2} \leq x_2 \\ & -x_1 \leq 1\end{array}$$

- (a) Calculate the optimal value and the set of optimal solutions.
- (b) Derive the dual function and calculate the dual optimal solution. Does strong duality hold? Why? [*Hint: show that $\lim_{x \rightarrow +\infty} (\sqrt{a + x^2} - x) = 0$ /*]

Solution 3:(a) For x_1 ,

$$\sqrt{x_1^2 + x_2^2} \leq x_2 \iff x_1^2 + x_2^2 \leq x_2^2 \iff x_1^2 \leq 0 \iff x_1 = 0$$

So the first constraint is equivalent to

$$x_1 = 0, \quad x_2 \geq 0.$$

The second constraint is

$$-x_1 \leq 1.$$

Substituting $x_1 = 0$, we get

$$0 \leq 1,$$

which is always satisfied. Therefore the feasible set is

$$\{(x_1, x_2) \mid x_1 = 0, x_2 \geq 0\}.$$

Since the objective is to minimize x_1 , and every feasible point satisfies $x_1 = 0$, the optimal value is

$$p^* = 0.$$

The set of optimal solutions is

$$X_{\text{opt}} = \{(0, t) \mid t \geq 0\}.$$

(b) Rewrite the constraints as

$$f_1(x) = \sqrt{x_1^2 + x_2^2} - x_2 \leq 0, \quad f_2(x) = -x_1 - 1 \leq 0.$$

Let the corresponding Lagrange multipliers be

$$\lambda \geq 0, \quad \nu \geq 0.$$

The Lagrangian is

$$L(x, \lambda, \nu) = x_1 + \lambda(\sqrt{x_1^2 + x_2^2} - x_2) + \nu(-x_1 - 1).$$

Rearranging terms gives

$$L(x, \lambda, \nu) = (1 - \nu)x_1 + \lambda(\sqrt{x_1^2 + x_2^2} - x_2) - \nu.$$

Therefore the dual function is

$$g(\lambda, \nu) = \inf_{x_1, x_2} L(x, \lambda, \nu) = -\nu + \inf_{x_1, x_2} \left[(1 - \nu)x_1 + \lambda(\sqrt{x_1^2 + x_2^2} - x_2) \right].$$

For fixed x_1 , consider

$$\sqrt{x_1^2 + x_2^2} - x_2.$$

Since

$$\sqrt{x_1^2 + x_2^2} \geq |x_2| \geq x_2,$$

we have

$$\sqrt{x_1^2 + x_2^2} - x_2 \geq 0.$$

On the other hand, using the hint with $a = x_1^2$, we obtain

$$\lim_{x_2 \rightarrow +\infty} \left(\sqrt{x_1^2 + x_2^2} - x_2 \right) = 0.$$

Hence

$$\inf_{x_2} \left(\sqrt{x_1^2 + x_2^2} - x_2 \right) = 0.$$

It follows that

$$g(\lambda, \nu) = -\nu + \inf_{x_1} (1 - \nu)x_1.$$

Now we distinguish two cases.

If $\nu = 1$, then

$$(1 - \nu)x_1 = 0,$$

so

$$g(\lambda, 1) = -1.$$

If $\nu \neq 1$, then $1 - \nu \neq 0$, and since x_1 is unrestricted in the infimum, we have

$$\inf_{x_1} (1 - \nu)x_1 = -\infty.$$

Therefore

$$g(\lambda, \nu) = -\infty.$$

Thus the dual function is

$$g(\lambda, \nu) = \begin{cases} -1, & \nu = 1, \lambda \geq 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

Hence the dual problem is

$$\max_{\lambda \geq 0, \nu \geq 0} g(\lambda, \nu),$$

and its optimal value is

$$d^* = -1.$$

The set of dual optimal solutions is

$$\{(\lambda, \nu) \mid \lambda \geq 0, \nu = 1\}.$$

Since in part (a) we found

$$p^* = 0,$$

while here

$$d^* = -1,$$

we have

$$d^* < p^*.$$

Therefore strong duality does *not* hold.

The reason is that Slater's condition fails. Indeed, for every (x_1, x_2) ,

$$\sqrt{x_1^2 + x_2^2} \geq |x_2| \geq x_2,$$

so the strict inequality

$$\sqrt{x_1^2 + x_2^2} < x_2$$

is impossible. Thus there is no strictly feasible point, and a duality gap occurs in this problem.

■

Exercise 4:

Consider the QP (over x_1, x_2):

$$\begin{aligned} & \text{minimize} && x_1^2 + 2x_2^2 - x_1x_2 - x_1 \\ & \text{subject to} && 5x_1 + 76x_2 \leq 1 \\ & && x_1 + 2x_2 \leq u_1 \\ & && x_1 - 4x_2 \leq u_2 \end{aligned}$$

- (a) Use CVXPY to find the primal/dual optimal values for $u_1 = -2$, $u_2 = -3$ and verify the KKT conditions (within reasonable numerical accuracy).
- (b) Use these values to predict the optimal value for

$$u_1 = -2 + \delta_1, \quad u_2 = -3 + \delta_2,$$

where

$$\delta_1, \delta_2 \in \{-0.1, 0.1\}$$

(four combinations). Verify that

$$p_{\text{pred}}^* \leq p_{\text{exact}}^*$$

and assess the prediction gap.

Solution 4:

Let

$$f(x_1, x_2) = x_1^2 + 2x_2^2 - x_1x_2 - x_1.$$

Then

$$\nabla f(x) = \begin{bmatrix} 2x_1 - x_2 - 1 \\ 4x_2 - x_1 \end{bmatrix}, \quad \nabla^2 f(x) = \begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix}.$$

Since

$$\det \begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix} = 8 - 1 = 7 > 0$$

and the leading principal minor is also positive, we have

$$\nabla^2 f(x) \succ 0.$$

Hence the objective is strictly convex.

(a) For

$$u_1 = -2, \quad u_2 = -3,$$

running `py/EX_4/4.py`, we obtain

$$x^* \approx \begin{bmatrix} -2.33333333 \\ 0.16666667 \end{bmatrix} = \begin{bmatrix} -\frac{7}{3} \\ \frac{1}{6} \end{bmatrix},$$

and

$$p^* = d^* = 8.22222222222221 \approx \frac{74}{9}.$$

so there is no duality gap numerically.

The dual multipliers are

$$\lambda^* \approx \begin{bmatrix} 0.04007173 \\ 2.74774125 \\ 2.88523345 \end{bmatrix}.$$

The three constraint slacks are

$$\begin{bmatrix} -5.32907052 \times 10^{-15} \\ -4.44089210 \times 10^{-16} \\ 0 \end{bmatrix},$$

so primal feasibility holds up to numerical precision. Since

$$\lambda^* \succeq 0,$$

dual feasibility also holds.

At the computed point,

$$\nabla f(x^*) \approx \begin{bmatrix} -5.83333333 \\ 3 \end{bmatrix}.$$

Let

$$A = \begin{bmatrix} 5 & 76 \\ 1 & 2 \\ 1 & -4 \end{bmatrix}.$$

Then the stationarity residual is

$$\nabla f(x^*) + A^T \lambda^* \approx \begin{bmatrix} 1.77635684 \times 10^{-15} \\ -1.77635684 \times 10^{-15} \end{bmatrix},$$

and

$$\|\nabla f(x^*) + A^T \lambda^*\|_\infty = 1.7763568394002505 \times 10^{-15}.$$

The complementary slackness residual is

$$\lambda_i^* g_i(x^*) \approx \begin{bmatrix} -2.13545062 \times 10^{-16} \\ -1.22024224 \times 10^{-15} \\ 0 \end{bmatrix},$$

and

$$\|\lambda_i^* g_i(x^*)\|_\infty = 1.2202422391705752 \times 10^{-15}.$$

Hence the KKT conditions hold within reasonable numerical accuracy.

Since this is a convex QP with affine constraints, and Slater's condition holds, strong duality holds.

The output is:

```
Base problem
status = optimal
x* = [-2.33333333  0.16666667]
p* = 8.222222222222221
d* = 8.222222222222221
primal-dual gap = 0.0
dual multipliers = [0.04007173  2.74774125  2.88523345]
slacks = [-5.32907052e-15 -4.44089210e-16  0.00000000e+00]

KKT check
gradient = [-5.83333333  3.          ]
stationarity residual = [ 1.77635684e-15 -1.77635684e-15]
||stationarity||_inf = 1.7763568394002505e-15
complementary slackness residual = [-2.13545062e-16 -1.22024224e-15
 0.00000000e+00]
||comp_slack||_inf = 1.2202422391705752e-15
dual feasibility (lam >= 0) = True
primal feasibility (slacks >= 0) = True
```

Scripts:

```
import cvxpy as cp
import numpy as np
from itertools import product
```

```
def solve_qp(u1, u2, solver=cp.OSQP):
    x = cp.Variable(2)

    P = np.array([
        [1.0, -0.5],
        [-0.5, 2.0]
    ])
    q = np.array([-1.0, 0.0])

    A = np.array([
        [5.0, 76.0],
        [1.0, 2.0],
        [1.0, -4.0]
    ])
    u = np.array([1.0, u1, u2])

    obj = cp.Minimize(cp.quad_form(x, P) + q @ x)

    c1 = A[0] @ x <= u[0]
    c2 = A[1] @ x <= u[1]
    c3 = A[2] @ x <= u[2]

    prob = cp.Problem(obj, [c1, c2, c3])
    prob.solve(solver=solver)

    xval = np.array(x.value, dtype=float).reshape(-1)
    lam = np.array([c1.dual_value, c2.dual_value, c3.dual_value],
                    dtype=float)

    slacks = u - A @ xval

    # Correct dual objective
    s = q + A.T @ lam
    dual_obj = -0.25 * s @ np.linalg.solve(P, s) - lam @ u

    return {
        "status": prob.status,
        "x": xval,
        "p_star": prob.value,
        "d_star": dual_obj,
        "duals": lam,
        "slacks": slacks,
        "A": A,
        "P": P,
        "q": q,
        "u": u
    }

# base point
u1_base, u2_base = -2, -3
res0 = solve_qp(u1_base, u2_base)

x_star = res0["x"]
```

```

p_star = res0["p_star"]
d_star = res0["d_star"]
lam = res0["duals"]
slacks = res0["slacks"]
A = res0["A"]

print("Base problem")
print("status =", res0["status"])
print("x* =", x_star)
print("p* =", p_star)
print("d* =", d_star)
print("primal-dual gap =", p_star - d_star)
print("dual multipliers =", lam)
print("slacks =", slacks)

# KKT check
grad = np.array([
    2*x_star[0] - x_star[1] - 1,
    4*x_star[1] - x_star[0]
], dtype=float)

stationarity = grad + A.T @ lam
comp_slack = lam * slacks

print("\nKKT check")
print("gradient =", grad)
print("stationarity residual =", stationarity)
print("||stationarity||_inf =", np.max(np.abs(stationarity)))
print("complementary slackness residual =", comp_slack)
print("||comp_slack||_inf =", np.max(np.abs(comp_slack)))
print("dual feasibility (lam >= 0) =", np.all(lam >= -1e-8))
print("primal feasibility (slacks >= 0) =", np.all(slacks >= -1e-8))
)

# prediction
print("\nPredictions")
for d1, d2 in product([-0.1, 0.1], repeat=2):
    u1 = -2 + d1
    u2 = -3 + d2
    pred = p_star - lam[1]*d1 - lam[2]*d2
    exact = solve_qp(u1, u2)["p_star"]
    print(f"delta=({d1:+.1f}, {d2:+.1f}) | pred={pred:.9f} | exact
          ={exact:.9f} | gap={exact-pred:.9f}")

```

(b) For

$$u_1 = -2 + \delta_1, \quad u_2 = -3 + \delta_2, \quad \delta_1, \delta_2 \in \{-0.1, 0.1\},$$

the sensitivity-based prediction is

$$p_{\text{pred}}^* = p^* - \lambda_2^* \delta_1 - \lambda_3^* \delta_2.$$

Using

$$p^* = 8.22222222222221, \quad \lambda_2^* = 2.74774125, \quad \lambda_3^* = 2.88523345,$$

we get

$$p_{\text{pred}}^* = 8.22222222222221 - 2.74774125 \delta_1 - 2.88523345 \delta_2.$$

Running `py/EX_4/4.py`, we obtain the following four cases:

δ_1	δ_2	p_{pred}^*	p_{exact}^*	$p_{\text{exact}}^* - p_{\text{pred}}^*$
-0.1	-0.1	8.785519692	8.815555556	0.030035864
-0.1	0.1	8.208473002	8.318888889	0.110415887
0.1	-0.1	8.235971442	8.706386719	0.470415276
0.1	0.1	7.658924753	7.751525608	0.092600855

Hence, in all four cases,

$$p_{\text{pred}}^* \leq p_{\text{exact}}^*.$$

So the prediction is indeed a valid lower bound.

The smallest prediction gap is

$$0.030035864$$

for $(\delta_1, \delta_2) = (-0.1, -0.1)$, while the largest prediction gap is

$$0.470415276$$

for $(\delta_1, \delta_2) = (0.1, -0.1)$. Therefore, the first-order prediction is accurate for some perturbation directions, but more conservative for others.

The output is:

```
Predictions
delta=(-0.1, -0.1) | pred=8.785519692 | exact=8.815555556 | gap
=0.030035864
delta=(-0.1, +0.1) | pred=8.208473002 | exact=8.318888889 | gap
=0.110415887
delta=(+0.1, -0.1) | pred=8.235971442 | exact=8.706386719 | gap
=0.470415276
delta=(+0.1, +0.1) | pred=7.658924753 | exact=7.751525608 | gap
=0.092600855
```



Exercise 5:

Derive the duals for problems a, b, solve them and verify the KKT conditions:

(a)

$$\begin{array}{ll}\text{minimize} & x_1 + 2x_2 + 3x_3 - x_4 \\ \text{subject to} & x_1 - x_2 + x_3 - 2x_4 \leq 6, \\ & 2x_1 + x_2 - x_3 \geq 2, \\ & -x_1 + x_2 - x_3 + x_4 \geq 8, \\ & 0 \leq x_1 \leq 3, \\ & 1 \leq x_2 \leq 4, \\ & 0 \leq x_3 \leq 10, \\ & 2 \leq x_4 \leq 5.\end{array}$$

(b)

$$\begin{array}{ll}\text{minimize} & 9x_1^2 + 9x_2^2 - 30x_1 - 72x_2 \\ \text{subject to} & -2x_1 - x_2 \geq -4, \\ & x_1 \geq 0, \ x_2 \geq 0.\end{array}$$

Solution 5:

Since both problems are convex and satisfy Slater's condition, strong duality holds, and the KKT conditions are necessary and sufficient.

(a) Let

$$c = (1, 2, 3, -1)^T, \quad a_1 = (1, -1, 1, -2)^T, \quad a_2 = (2, 1, -1, 0)^T, \quad a_3 = (-1, 1, -1, 1)^T,$$

and

$$\ell = (0, 1, 0, 2)^T, \quad u = (3, 4, 10, 5)^T.$$

Then the primal problem is

$$\min c^T x$$

subject to

$$a_1^T x \leq 6, \quad a_2^T x \geq 2, \quad a_3^T x \geq 8, \quad \ell \leq x \leq u.$$

We rewrite all inequalities in the form " ≤ 0 ":

$$a_1^T x - 6 \leq 0, \quad 2 - a_2^T x \leq 0, \quad 8 - a_3^T x \leq 0,$$

$$\ell - x \leq 0, \quad x - u \leq 0.$$

Introduce multipliers

$$y_1, y_2, y_3 \geq 0, \quad \alpha \geq 0, \quad \beta \geq 0.$$

The Lagrangian is

$$L(x, y, \alpha, \beta) = c^T x + y_1(a_1^T x - 6) + y_2(2 - a_2^T x) + y_3(8 - a_3^T x) + \alpha^T(\ell - x) + \beta^T(x - u).$$

Hence

$$L = (c + y_1 a_1 - y_2 a_2 - y_3 a_3 - \alpha + \beta)^T x - 6y_1 + 2y_2 + 8y_3 + \ell^T \alpha - u^T \beta.$$

Therefore, the dual function is finite iff

$$c + y_1 a_1 - y_2 a_2 - y_3 a_3 - \alpha + \beta = 0.$$

So the dual problem is

$$\begin{aligned} \max \quad & -6y_1 + 2y_2 + 8y_3 + \ell^T \alpha - u^T \beta \\ \text{s.t.} \quad & c + y_1 a_1 - y_2 a_2 - y_3 a_3 - \alpha + \beta = 0, \\ & y \geq 0, \alpha \geq 0, \beta \geq 0. \end{aligned}$$

Expanding the equality constraints gives

$$\begin{aligned} 1 + y_1 - 2y_2 + y_3 - \alpha_1 + \beta_1 &= 0, \\ 2 - y_1 - y_2 - y_3 - \alpha_2 + \beta_2 &= 0, \\ 3 + y_1 + y_2 + y_3 - \alpha_3 + \beta_3 &= 0, \\ -1 - 2y_1 - y_3 - \alpha_4 + \beta_4 &= 0. \end{aligned}$$

The dual objective becomes

$$-6y_1 + 2y_2 + 8y_3 + \alpha_2 + 2\alpha_4 - 3\beta_1 - 4\beta_2 - 10\beta_3 - 5\beta_4.$$

Now solve the primal. Since the objective is

$$x_1 + 2x_2 + 3x_3 - x_4,$$

we want x_4 as large as possible and x_1, x_2, x_3 as small as possible. Take

$$x_4 = 5, \quad x_1 = 0, \quad x_3 = 0.$$

Then the third constraint becomes

$$-x_1 + x_2 - x_3 + x_4 \geq 8 \implies x_2 + 5 \geq 8,$$

so $x_2 \geq 3$. To minimize the objective we take $x_2 = 3$. Thus

$$x^* = (0, 3, 0, 5)^T, \quad p^* = 0 + 2 \cdot 3 + 0 - 5 = 1.$$

Therefore, according to slackness and stationarity, a feasible dual point is

$$y^* = (0, 0, 2), \quad \alpha^* = (3, 0, 5, 0), \quad \beta^* = (0, 0, 0, 3).$$

Indeed,

$$c + y_1 a_1 - y_2 a_2 - y_3 a_3 - \alpha + \beta = (1, 2, 3, -1) - 2(-1, 1, -1, 1) - (3, 0, 5, 0) + (0, 0, 0, 3) = 0.$$

The dual value is

$$d^* = -6 \cdot 0 + 2 \cdot 0 + 8 \cdot 2 + \ell^T \alpha^* - u^T \beta^* = 16 - 15 = 1.$$

Hence

$$d^* = p^* = 1,$$

so y^*, α^*, β^* is dual optimal.

Now verify the KKT conditions.

Primal feasibility:

$$x^* = (0, 3, 0, 5)^T$$

satisfies all primal constraints.

Dual feasibility:

$$y^* \geq 0, \quad \alpha^* \geq 0, \quad \beta^* \geq 0.$$

Complementary slackness:

$$y_1(a_1^T x^* - 6) = 0, \quad y_2(2 - a_2^T x^*) = 0, \quad y_3(8 - a_3^T x^*) = 2(8 - 8) = 0,$$

and

$$\alpha_i(\ell_i - x_i^*) = 0, \quad \beta_i(x_i^* - u_i) = 0, \quad i = 1, 2, 3, 4.$$

In particular,

$$\alpha_1(0 - 0) = 0, \quad \alpha_3(0 - 0) = 0, \quad \beta_4(5 - 5) = 0.$$

Stationarity:

$$c + y_1 a_1 - y_2 a_2 - y_3 a_3 - \alpha + \beta = 0,$$

which we already checked explicitly above.

Therefore all KKT conditions hold.

(b) Rewrite the constraint

$$-2x_1 - x_2 \geq -4$$

as

$$2x_1 + x_2 \leq 4.$$

So the primal problem is

$$\begin{aligned} \min \quad & 9x_1^2 + 9x_2^2 - 30x_1 - 72x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 - 4 \leq 0, \\ & -x_1 \leq 0, \quad -x_2 \leq 0. \end{aligned}$$

Introduce multipliers

$$\lambda \geq 0, \quad \mu_1 \geq 0, \quad \mu_2 \geq 0.$$

The Lagrangian is

$$L(x, \lambda, \mu_1, \mu_2) = 9x_1^2 + 9x_2^2 - 30x_1 - 72x_2 + \lambda(2x_1 + x_2 - 4) - \mu_1 x_1 - \mu_2 x_2.$$

Thus

$$L = 9x_1^2 + (2\lambda - 30 - \mu_1)x_1 + 9x_2^2 + (\lambda - 72 - \mu_2)x_2 - 4\lambda.$$

To compute the dual function, minimize with respect to x_1, x_2 :

$$\frac{\partial L}{\partial x_1} = 18x_1 + 2\lambda - 30 - \mu_1 = 0, \quad \frac{\partial L}{\partial x_2} = 18x_2 + \lambda - 72 - \mu_2 = 0.$$

Hence

$$x_1 = \frac{30 + \mu_1 - 2\lambda}{18}, \quad x_2 = \frac{72 + \mu_2 - \lambda}{18}.$$

Substituting back gives

$$g(\lambda, \mu_1, \mu_2) = -4\lambda - \frac{(2\lambda - 30 - \mu_1)^2}{36} - \frac{(\lambda - 72 - \mu_2)^2}{36}.$$

Therefore the dual problem is

$$\max_{\lambda, \mu_1, \mu_2 \geq 0} \left[-4\lambda - \frac{(2\lambda - 30 - \mu_1)^2}{36} - \frac{(\lambda - 72 - \mu_2)^2}{36} \right].$$

Now solve the primal. The unconstrained minimizer satisfies

$$18x_1 - 30 = 0, \quad 18x_2 - 72 = 0,$$

so

$$x_1 = \frac{5}{3}, \quad x_2 = 4.$$

This is infeasible since

$$2 \cdot \frac{5}{3} + 4 = \frac{22}{3} > 4.$$

Hence the optimum lies on the active boundary

$$2x_1 + x_2 = 4.$$

Let

$$x_2 = 4 - 2x_1, \quad 0 \leq x_1 \leq 2.$$

Then

$$\phi(x_1) = 9x_1^2 + 9(4 - 2x_1)^2 - 30x_1 - 72(4 - 2x_1) = 45x_1^2 - 30x_1 - 144.$$

Differentiate:

$$\phi'(x_1) = 90x_1 - 30.$$

Setting $\phi'(x_1) = 0$ gives

$$x_1^* = \frac{1}{3}, \quad x_2^* = 4 - 2 \cdot \frac{1}{3} = \frac{10}{3}.$$

Therefore

$$x^* = \left(\frac{1}{3}, \frac{10}{3} \right)^T.$$

The optimal value is

$$p^* = 9 \left(\frac{1}{3} \right)^2 + 9 \left(\frac{10}{3} \right)^2 - 30 \left(\frac{1}{3} \right) - 72 \left(\frac{10}{3} \right) = 1 + 100 - 10 - 240 = -149.$$

Now solve the dual. Since $x_1^* > 0$ and $x_2^* > 0$, complementary slackness gives

$$\mu_1^* = 0, \quad \mu_2^* = 0.$$

Using stationarity,

$$18x_1 - 30 + 2\lambda - \mu_1 = 0, \quad 18x_2 - 72 + \lambda - \mu_2 = 0.$$

Substitute $x^* = (1/3, 10/3)$ and $\mu_1^* = \mu_2^* = 0$:

$$18 \cdot \frac{1}{3} - 30 + 2\lambda = 0 \implies 6 - 30 + 2\lambda = 0 \implies \lambda^* = 12.$$

So the dual optimal point is

$$(\lambda^*, \mu_1^*, \mu_2^*) = (12, 0, 0).$$

Its value is

$$d^* = -4 \cdot 12 - \frac{(24 - 30)^2}{36} - \frac{(12 - 72)^2}{36} = -48 - 1 - 100 = -149.$$

Hence

$$d^* = p^* = -149.$$

Finally, verify KKT.

Primal feasibility:

$$2x_1^* + x_2^* = 2 \cdot \frac{1}{3} + \frac{10}{3} = 4, \quad x_1^* > 0, \quad x_2^* > 0.$$

Dual feasibility:

$$\lambda^* = 12 \geq 0, \quad \mu_1^* = \mu_2^* = 0.$$

Complementary slackness:

$$\lambda^*(2x_1^* + x_2^* - 4) = 12 \cdot 0 = 0,$$

$$\mu_1^*x_1^* = 0, \quad \mu_2^*x_2^* = 0.$$

Stationarity:

$$18x_1^* - 30 + 2\lambda^* - \mu_1^* = 18 \cdot \frac{1}{3} - 30 + 24 = 6 - 30 + 24 = 0,$$

$$18x_2^* - 72 + \lambda^* - \mu_2^* = 18 \cdot \frac{10}{3} - 72 + 12 = 60 - 72 + 12 = 0.$$

Therefore all KKT conditions hold.

■

Exercise 6:

The following problem is for *extra credit*.

Consider the projection of a point on the ℓ_1 unit ball:

$$\begin{aligned} & \text{minimize} && \frac{1}{2} \|x - a\|_2^2 \\ & \text{subject to} && \|x\|_1 \leq 1 \end{aligned}$$

Derive an efficient method for the dual problem and show how to use it to solve the primal. Compare the run-time of your method vs. a CVXPY solution for randomly chosen $a \in \mathbb{R}^{300}$.

[Hint: Show the dual function is differentiable and consider two cases: a) $\|a\|_1 \leq 1$ and b) $\|a\|_1 > 1$]

Solution 6:

Let

$$\begin{aligned} & \text{minimize} \quad \frac{1}{2}\|x - a\|_2^2 \\ & \text{subject to} \quad \|x\|_1 \leq 1. \end{aligned}$$

This is the Euclidean projection of a onto the ℓ_1 unit ball. Since $x = 0$ satisfies $\|0\|_1 = 0 < 1$, Slater's condition holds, so strong duality applies.

Introduce the multiplier $\lambda \geq 0$ for the constraint $\|x\|_1 - 1 \leq 0$. The Lagrangian is

$$L(x, \lambda) = \frac{1}{2}\|x - a\|_2^2 + \lambda(\|x\|_1 - 1).$$

Hence the dual function is

$$g(\lambda) = \inf_x L(x, \lambda) = \inf_x \left(\frac{1}{2}\|x - a\|_2^2 + \lambda\|x\|_1 \right) - \lambda, \quad \lambda \geq 0.$$

Because the objective is separable across coordinates,

$$g(\lambda) = -\lambda + \sum_{i=1}^n \inf_{x_i} \left[\frac{1}{2}(x_i - a_i)^2 + \lambda|x_i| \right].$$

For each coordinate, consider

$$\min_{x_i} \frac{1}{2}(x_i - a_i)^2 + \lambda|x_i|.$$

Its minimizer is the soft-thresholding operator

$$x_i(\lambda) = \text{sign}(a_i)(|a_i| - \lambda)_+.$$

Therefore,

$$x(\lambda) = S_\lambda(a).$$

Substituting this minimizer back gives

$$g(\lambda) = -\lambda + \sum_{i=1}^n \phi_i(\lambda),$$

where

$$\phi_i(\lambda) = \begin{cases} \frac{1}{2}a_i^2, & |a_i| \leq \lambda, \\ \lambda|a_i| - \frac{1}{2}\lambda^2, & |a_i| > \lambda. \end{cases}$$

For each i ,

$$\phi'_i(\lambda) = \begin{cases} 0, & |a_i| \leq \lambda, \\ |a_i| - \lambda, & |a_i| > \lambda. \end{cases}$$

At $\lambda = |a_i|$, both one-sided derivatives are 0, so each ϕ_i is differentiable. Hence g is differentiable, with

$$g'(\lambda) = -1 + \sum_{i=1}^n (|a_i| - \lambda)_+ = \|S_\lambda(a)\|_1 - 1.$$

Since g is a dual function, it is concave. Therefore the dual problem is the one-dimensional concave maximization problem

$$\max_{\lambda \geq 0} g(\lambda).$$

Next we consider the two cases in the hint.

Case 1: $\|a\|_1 \leq 1$. Then a is already feasible, and the objective $\frac{1}{2}\|x - a\|_2^2$ is minimized at $x = a$. Thus

$$x^* = a, \quad \lambda^* = 0.$$

Case 2: $\|a\|_1 > 1$. Then the optimum lies on the boundary $\|x^*\|_1 = 1$. Since

$$x^* = S_{\lambda^*}(a),$$

the optimal dual variable satisfies

$$g'(\lambda^*) = 0 \iff \sum_{i=1}^n (|a_i| - \lambda^*)_+ = 1.$$

Hence the primal solution is recovered from the dual solution by

$$x_i^* = \text{sign}(a_i) \max(|a_i| - \lambda^*, 0), \quad i = 1, \dots, n.$$

To compute λ^* efficiently, let

$$u_1 \geq u_2 \geq \dots \geq u_n$$

be the entries of $|a|$ sorted in decreasing order, and define

$$s_k = \sum_{i=1}^k u_i.$$

If exactly the first ρ entries are active, then

$$\sum_{i=1}^{\rho} (u_i - \lambda^*) = 1,$$

so

$$\lambda^* = \frac{s_\rho - 1}{\rho}.$$

The correct index is

$$\rho = \max \left\{ k \in \{1, \dots, n\} : u_k - \frac{s_k - 1}{k} > 0 \right\}.$$

Thus the algorithm is:

```

if  $\|a\|_1 \leq 1$ , return  $x^* = a$ ,
otherwise sort  $|a|$  into  $u_1 \geq \dots \geq u_n$ ,
 $\rho = \max \left\{ k : u_k - \frac{s_k - 1}{k} > 0 \right\}$ ,
 $\lambda^* = \frac{s_\rho - 1}{\rho}$ ,
 $x^* = \text{sign}(a) \odot \max(|a| - \lambda^*, 0)$ .

```

Its complexity is dominated by sorting, so it is $O(n \log n)$.

For a random vector $a \in \mathbb{R}^{300}$, the numerical results are

custom time = 0.0012628999538719654, cvxpy time = 0.02643220010213554,

$$\max_i |x_i^{\text{custom}} - x_i^{\text{cvx}}| = 6.824176490036887 \times 10^{-6},$$

$$f(x^{\text{custom}}) = 147.8812856381681, \quad f(x^{\text{cvx}}) = 147.8812857727218,$$

$$\lambda^* = 2.2569512699145116.$$

Therefore the custom method and the CVXPY solution agree up to numerical precision, while the custom method is faster by a factor of about

$$\frac{0.02643220010213554}{0.0012628999538719654} \approx 20.93.$$

Since $\lambda^* > 0$, this experiment belongs to the case $\|a\|_1 > 1$, and the optimal primal solution is obtained by soft-thresholding.

The output is:

```

custom time: 0.0012628999538719654
cvxpy time: 0.02643220010213554
max abs diff: 6.824176490036887e-06
custom objective: 147.8812856381681
cvx objective: 147.8812857727218
lambda*: 2.2569512699145116

```

Scripts:

```

import numpy as np
import time
import cvxpy as cp

def proj_l1_ball(a, radius=1.0):
    a = np.asarray(a, dtype=float)
    if np.linalg.norm(a, 1) <= radius:
        return a.copy(), 0.0

    u = np.sort(np.abs(a))[:, -1]
    cssv = np.cumsum(u)

```

```
rho = np.nonzero(u - (cssv - radius) / np.arange(1, len(u) + 1) > 0)
      [0][-1] + 1
lam = (cssv[rho - 1] - radius) / rho
x = np.sign(a) * np.maximum(np.abs(a) - lam, 0.0)
return x, lam

# random test
np.random.seed(0)
n = 300
a = np.random.randn(n)

# custom method
t0 = time.perf_counter()
x_custom, lam = proj_l1_ball(a)
t1 = time.perf_counter()

# CVXPY
x = cp.Variable(n)
prob = cp.Problem(cp.Minimize(0.5 * cp.sum_squares(x - a)),
                  [cp.norm1(x) <= 1])

t2 = time.perf_counter()
val = prob.solve(solver=cp.CLARABEL, verbose=False)
t3 = time.perf_counter()

x_cvx = np.array(x.value).reshape(-1)

print("custom time:", t1 - t0)
print("cvxpy time:", t3 - t2)
print("max abs diff:", np.max(np.abs(x_custom - x_cvx)))
print("custom objective:", 0.5 * np.sum((x_custom - a) ** 2))
print("cvx objective:", 0.5 * np.sum((x_cvx - a) ** 2))
print("lambda*:", lam)
```

